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Exploring the complexity of urban walking environments based on pedestrian navigation data

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Abstract

The complexity of urban environments influences pedestrians' walkability, which is especially significant for people living in mega cities. While many studies identify influential factors, how these factors shape pedestrian wayfinding through complex and spatially varied mechanisms remains underexplored. This study addresses this gap by using a novel pedestrian navigation dataset as a proxy to quantify the perceived complexity of walking environments. By integrating multi-scale urban features—four at the macro-level and 14 at the micro-level derived from Street View Imagery—we systematically uncover the key correlates of navigation demand and their underlying effects. The results reveal that a combination of factors such as the number of Points of Interest, transportation accessibility, proportion of people in view, and intersection count are positively associated with pedestrians' navigation behavior. More importantly, we demonstrate that their relationship is profoundly non-linear and exhibits strong spatial heterogeneity. These results are further validated through population normalization, sensitivity tests, and temporal comparisons between weekdays and weekends. Such analyses confirm the robust and independent association between environmental complexity and navigation behavior. By operationalizing these complex interrelationships, our work advances the theoretical framework for urban environmental complexity. The findings provide crucial evidence for moving beyond a "one-size-fits-all" approach, offering targeted, context-aware insights to foster truly human-centered urban planning and design.

Keywords Urban complexity, Navigation demand, Pedestrian wayfinding, Multi-scale analysis, GWRF, SHAP

1 Introduction

The complexity of urban spatial configurations is increasing as a result of the growth in mega city development, impacting pedestrians' walkability in cities (Chen et al., 2025). Distinctive urban landscapes and intuitive urban design are crucial to better understand the spatial structures that can help people reach their destinations effortlessly (Ramanoël et al., 2022). However, in environments lacking such spatial cues, individuals tend to rely on tools such as map applications for navigation. Intuitive environmental perception should thus be emphasized in urban planning, as it plays a critical role in facilitating easy navigation through unfamiliar environments and ensuring the development of genuinely livable pedestrian areas (Li, J. et al., 2025).

The study of human experience within urban settings fundamentally concerns the concept of "place"—a spatial location imbued with meaning through human interaction. Foundational theories posit that individuals navigate cities by forming "cognitive maps" or mental representations of the spatial layout (De Tinguy et al., 2024). A city's "legibility"—the ease with which its parts can be recognized and organized into a coherent pattern—is crucial for effective wayfinding (Ahmadpoor & Smith, 2020).

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Environments lacking coherent structures or clear landmarks can disrupt this cognitive mapping process, leading to disorientation, spatial anxiety, and high cognitive loads on individuals (Choi et al., 2023). Specifically, prior research has identified environmental attributes as critical determinants of navigational performance (Jamshidi et al., 2020). Key factors contributing to wayfinding difficulty include irregular street layouts, the lack of distinct landmarks, and high levels of visual clutter.

The field of urban studies introduced the concept of "walkability" to operationalize these ideas and assess the quality of the pedestrian realm (He & He, 2023; Moura et al., 2017). Modern walkability frameworks have evolved beyond simple metrics, proposing comprehensive assessments based on hierarchical pedestrian needs such as safety, convenience, continuity, and attractiveness (Liu et al., 2023). Although these frameworks are instrumental in evaluating the overall quality of the walking environment, they primarily aim to create a composite index for public health (Jiang et al., 2022) and environmental sustainability (Zhou et al., 2024). Consequently, they may not sufficiently delineate the specific cognitive factors that precipitate wayfinding difficulties. Thus, a neighborhood can be classified as highly "walkable" based on its amenities and safety, despite simultaneously presenting considerable perceptual complexities and wayfinding challenges, particularly for individuals unfamiliar with the area (Celik & Yildirim, 2025). Studies investigating these more nuanced dimensions of environmental perception have traditionally relied on subjective data collection methods such as field surveys, interviews, and manual audits. Although more advanced neuroscientific tools like electroencephalogram (EEG) and functional magnetic resonance imaging (fMRI) have also been employed (Makanadar, 2024), these approaches, whether survey- or lab-based, exhibit limited scalability and are often ill-suited for capturing perceptual dynamics across large-scale and/or heterogeneous urban contexts.

To overcome these limitations, a new research paradigm that leverages large-scale behavioral data to infer human–environment interactions has emerged. Although foundational data sources such as traffic flows (Zhang, F. 2018) and cycling patterns (Wang, L. et al., 2023) are valuable for modeling urban dynamics, they fail to focus on pedestrian navigational cognitions. Other trajectory datasets have similar constraints. For instance, vehicular data (e.g., floating car data) reveal macro-scale traffic patterns but are ill-suited for capturing pedestrian-scale experiences (Jiao et al., 2022). Despite the ability of aggregated mobility data (e.g., cell tower records) to outline regional commuting networks (Zhang, W. et al., 2020), they are incapable of isolating navigation-specific behaviors. Moreover, although social media check-ins provide rich semantic context, they are constrained by demographic biases and discontinuous tracking (Shen & Karimi, 2016; Shen & Wu, 2022). Mobile navigation trajectories, conversely, present a unique opportunity to track continuous pedestrian journeys with high precision (Calabrese et al., 2013). These trajectories, captured precisely at moments when individuals actively seek directional assistance, can serve as a proxy for quantifying perceived urban complexity (Kong et al., 2018; Wang, D. et al., 2020).

The use of navigation trajectory counts as a proxy rests on cognitive wayfinding theory (Tahir & Krogstie, 2023): spatially complex environments that lack clear structure or distinctive landmarks induce elevated cognitive load, prompting individuals to seek navigation assistance. While individual motivations vary—including both genuine disorientation and habitual app use—we hypothesize that aggregate spatial patterns predominantly reflect environmental complexity. To validate this assumption, we conduct comprehensive robustness checks including population normalization, temporal consistency tests, and sensitivity analyses (Supplementary Material S2-S4).

Building on this theoretical foundation, this study tests the hypothesis that the count of pedestrian navigation trajectories serves as a quantifiable proxy for "navigational demand" (Ahmadpoor & Smith, 2020; Bongiorno et al., 2021). This demand is interpreted as a reflection of the "perceived navigational complexity" of the environment—a measure of the cognitive load required for wayfinding. This approach marks a shift from analyzing abstract "walkability" to modeling the micro-scale influence of specific environmental features on real-time navigation behavior. Unlike aggregate pedestrian flows, which measure overall footfall, navigation trajectories may capture moments of uncertainty and decision making, offering a valuable lens through which to examine the cognitive demands of wayfinding (Zhao et al., 2025).

From the perspective of urban complexity, explaining the environmental features that precipitate wayfinding challenges requires a sophisticated and multi-scale measurement framework. Historically, urban research has relied on macro-scale approaches, such as the foundational "3Ds" (density, diversity, design) (Cervero & Kockelman, 1997) and the extended "5Ds" (distance and destinations included) (Ewing & Cervero, 2010), which established key variables such as land-use mix and intersection connectivity. However, despite their value, these metrics often fail to capture the nuanced, eye-level interactions that directly shape a pedestrian's perceptual experience. To bridge this gap, recent studies have increasingly integrated micro-scale analysis—a shift propelled by the availability of novel data sources such as Street View Imagery (SVI) (Li, Y. et al., 2022).

The application of machine learning to SVI enables the systematic extraction of crucial visual features, including green space visibility and street enclosure (Biljecki & Ito, 2021; Yang, W. et al., 2023). Such integration has spurred the development of more holistic models, notably the "visual-structural-semantic" framework (Guan et al., 2022)—an integrated macro–micro approach that simultaneously considers visual entropy (a micro-scale perceptual experience), intersection patterns (a macro-scale spatial structure), and Points of Interest (POI) metrics (a sociocultural context). Continuous advancements to such frameworks provide a robust foundation to quantitatively link the urban environment's multi-scale features to the nuanced realities of human spatial perception—a methodological approach this study adopts.

Therefore, this study uses the pedestrian navigation trajectory data to quantify the perceived complexity and cognitive load imposed by the urban environment. The most critical environmental determinants are then uncovered from a multitude of factors. Subsequently, the complex and spatially varied mechanisms through which they operate are analyzed. This research aims to answer the following research questions (RQs):

> RQ1: What are the primary macro- and micro-scale urban environmental features associated with the intensity of pedestrian navigation demand?
> RQ2: How are these key factors associated with pedestrian navigation demand through complex mechanisms, including non-linear effects and spatially heterogeneous impacts?

2 Study area and data sources

2.1 Study area

Shanghai, situated on the alluvial plain of the Yangtze River Delta in eastern China, is a designated national central city and a key hub for the country's economy, finance, and technological innovation (Fig. 1a). Its complex urban spatial structure and diverse socioeconomic characteristics make it a representative case for studying a typical national central city. Shanghai, characterized by high population density and economic vitality, covers a total area of 6,340.5 square kilometers with a resident population exceeding 20 million as of 2022.

The study area (Fig. 1b) encompasses the main island of Shanghai, excluding Chongming District because of its geographical isolation and lack of spatial continuity with the main urban area. To provide an analytical context, the city is broadly composed of a high-density city center (core urban area) and surrounding suburban areas. The core comprises seven administrative districts: Huangpu, Xuhui, Jing'an, Changning, Yangpu, Hongkou, and Putuo. The primary unit of analysis in this research is the traffic analysis zone (TAZ), which is derived from the road network. The entire study area is subdivided into these TAZs (Fig. 1c) that provide the essential spatial framework for aggregating pedestrian navigation data and other environmental features for subsequent modeling.

2.2 Data sources

The data used in this study comprise five categories: pedestrian navigation data; SVI, POI, and Area of Interest (AOI) data; road network data; and population grid data. These multi-source datasets collectively form the foundation for analyzing urban complexity.

The navigation dataset—covering the period from September 16 to 19, 2017—was obtained from a mobile navigation service provider in Shanghai. To ensure privacy, all user identifiers were anonymized. The raw data, sampled at one-second intervals, includes fields such as geographic coordinates, instantaneous velocity, and bearing. Although the data encompasses various modes of transportation, we retained only pedestrian data to mitigate potential biases from non-exploratory uses (e.g., commuting or personal habits). Although individual navigation behaviors may be influenced by factors such as habits or estimated time of arrival queries, our analysis focuses on macroscopic spatial patterns aggregated from millions of trajectories, where individual "noise" tends to distribute randomly, allowing systematic "signals" driven by urban features (e.g., road networks, functional layouts) to emerge prominently.

SVI was sourced from the Baidu API (https://map.baidu.com) in September 2017. From an initial uniform sampling designed to cover the entire city of Shanghai, a final dataset of 198,156 valid points was obtained after filtering for data availability. Each point comprises four high-resolution images captured at 0°, 90°, 180°, and 270° viewing angles, collectively providing a panoramic representation of the street environment.

POI, AOI, and metro infrastructure data were sourced from Amap (www.amap.com). This dataset, corresponding to the year 2017, is used to analyze key macro-level indicators. Specifically, the categorization and location of POIs and AOIs are essential for calculating the land use entropy and the raw POI count within each analysis zone. Furthermore, the geographic coordinates of metro stations serve as fundamental supply points in the transportation accessibility analysis (Long et al., 2024).

The road network was obtained from OpenStreetMap (www.openstreetmap.org) for the year 2017. This dataset provides the structural backbone of the urban form, from which the intersection variable was derived by quantifying the number of road junctions within each grid cell.

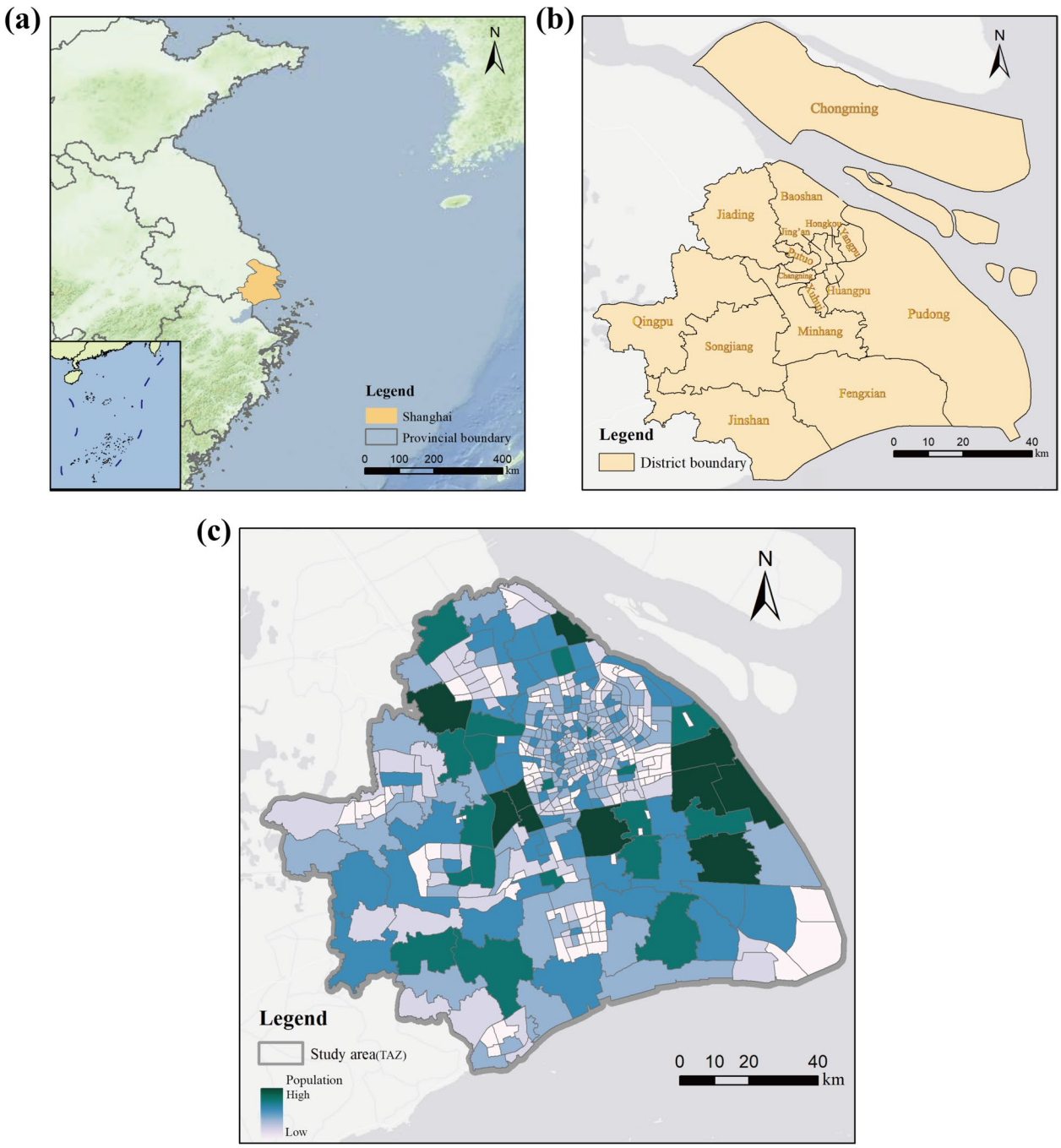


Fig. 1 Study area

This serves as a direct measure of street connectivity, a critical component of urban design (Hosseini et al., 2025).

A 100 m-resolution population grid was sourced from WorldPop (https://hub.worldpop.org). This dataset provides fine-grained estimates of residential population distribution in 2017, serving as a crucial input for weighting the demand side in the transportation accessibility model (Fan et al., 2017). It enables a more accurate representation of the potential population served by public transit within each TAZ.

2.3 Data preprocessing

Prior to analysis, all datasets were preprocessed to ensure quality and consistency. Key procedures include

cleaning, filtering, map-matching the raw pedestrian navigation trajectories, and quality control for the SVI.

The pedestrian navigation dataset necessitates a comprehensive preprocessing procedure (Fig. 2). Initial steps include duplicate removal, outlier filtering, and reordering the data based on user IDs and timestamps. To isolate pedestrian trajectories, a set of motion features, such as angular velocity and acceleration, are computed Xiao, Z. et al., 2017). These features are then used to train a LightGBM classifier on labeled GeoLife samples (Yu, 2021), achieving an accuracy of 93.8% in distinguishing among pedestrian, bicycle, and motor vehicle modes (Wu et al., 2022). Only trajectory points classified as pedestrian were retained for further analysis.

To reduce computational load, these trajectories were temporally segmented into three-minute intervals and then grouped by user ID to extract Origin–Destination (OD) points. The resulting OD records were further filtered to exclude trips with durations longer than 7,200 s. Finally, a nearest-neighbor method was applied to match trajectory origin points to the road network, correcting for Global Positioning System (GPS) drift (Lan et al., 2020). These procedures ultimately generated 3,741,976 valid navigation records for analysis. Table 1 presents an example of this data, which can subsequently be used directly to accumulate pedestrian navigation OD trajectory points.

For SVI data, a rigorous quality control process was implemented (Fig. 3), excluding images with severe

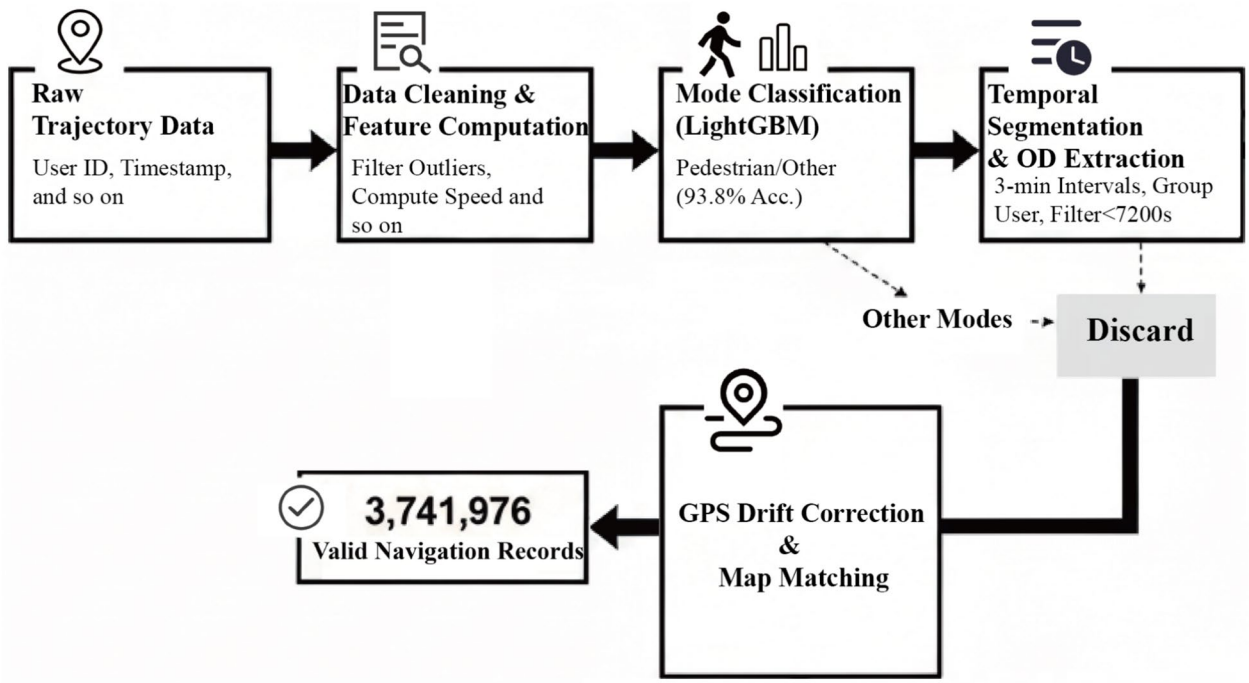


Fig. 2 Pedestrian navigation data preprocessing

Table 1 Description of the pedestrian navigation trajectory

UserID	St_lon	St_lat	St_time	Ed_lon	Ed_lat	Ed_time
a92b55699f7e7834	121.244144	31.019407	2017–09–16 09:46:40	121.243752	31.017681	2017–09–16 09:48:34
a8b857838371b7f5	121.451529	31.365481	2017–09–16 15:06:57	121.445944	31.358530	2017–09–16 15:15:35
a53627352374aec2	121.338355	31.257870	2017–09–16 21:49:29	121.338176	31.258210	2017–09–16 21:50:13
aab52f2b71f8f22c	121.521303	31.280951	2017–09–16 09:38:36	121.521173	31.281010	2017–09–16 09:39:08
afa810631c6d414d	121.331948	31.150164	2017–09–16 13:11:44	121.334446	31.151506	2017–09–16 13:13:17

UserID is the user ID of the trajectory, which has been encrypted. St_lon and St_lat represent the longitude and latitude of the starting point of the trajectory, respectively. St_time represents the time of the start of the trajectory, Ed_lon represents the longitude of the endpoint of the trajectory, Ed_lat represents the dimension of the endpoint of the trajectory, Ed_time represents the time of the end of the trajectory

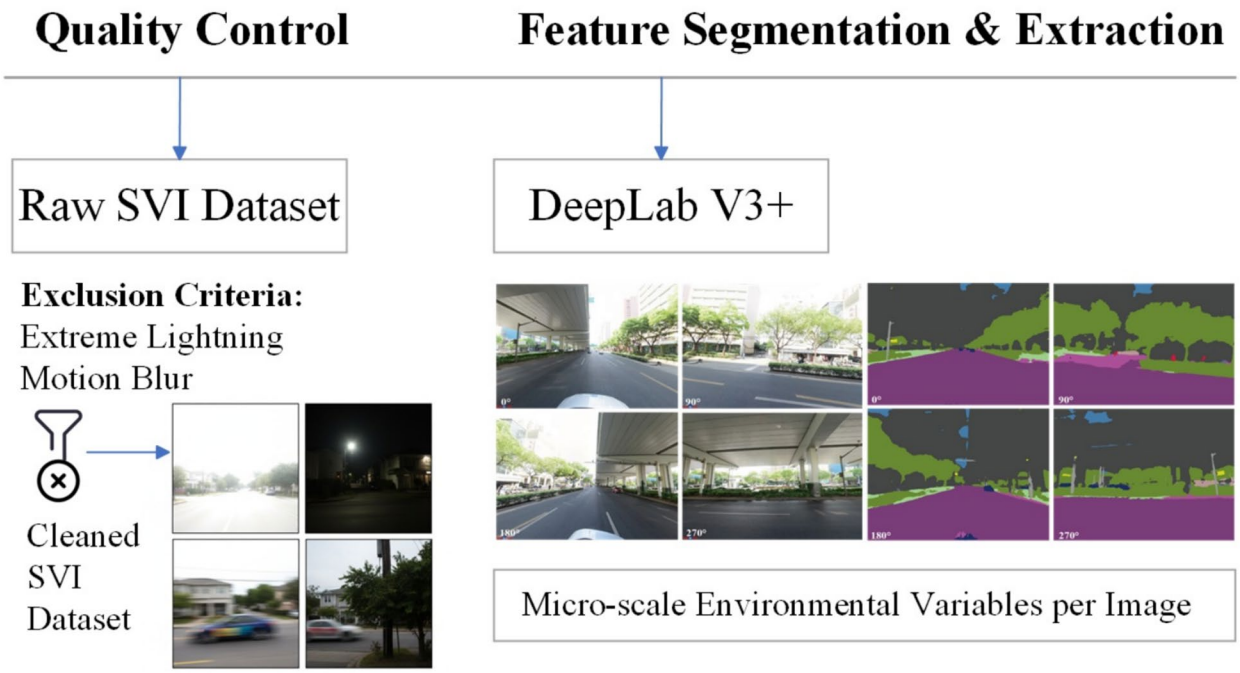


Fig. 3 SVI sampling and semantic segmentation

occlusion (exceeding 40%), extreme lighting conditions, or obvious motion blur to ensure dataset reliability. Semantic segmentation was then performed using the DeepLab V3+deep learning framework, identifying 19 key urban elements—including sky, vegetation, roads, and vehicles. For each image, pixel proportions corresponding to each semantic category were calculated to facilitate the subsequent extraction of micro-scale variables.

3 Methodology

This research employs a two-stage analytical framework to investigate the complex mechanisms underlying urban pedestrian navigation. As depicted in the research flowchart (Fig. 4), the framework comprises two core components: multi-scale variable construction and analytical modeling. The first stage quantifies pedestrian navigation demand and constructs 18 environmental features from both macro and micro perspectives. The second stage utilizes machine learning methods to analyze the relationships between these variables.

3.1 Variables

3.1.1 Dependent variable

Pedestrian navigation demand, quantified as the cumulative count of navigation origin points within each TAZ, was chosen as the dependent variable because it serves as a direct proxy for pedestrian activity and environmental unfamiliarity exposure (Millward, 2018). A higher frequency of navigation requests can indicate a greater need for wayfinding in non-routine travel.

3.1.2 Independent variables

The independent variables were selected to holistically measure the urban environment from two complementary perspectives. Macro-scale features capture the city's broader structural and functional organization, whereas micro-scale features represent the immediate eye-level perceptual experience of pedestrians.

The four macro-scale variables in Table 2 were selected to represent the city's foundational organization and functional meaning, corresponding to the structural and semantic dimensions of urban complexity (Guan et al., 2022). The selection is guided by the seminal "5Ds" (Zhan et al., 2025) framework, which identifies density, diversity, design, destination accessibility, and distance to transit as core urban environment attributes influencing travel behavior. Each of our macro variables maps directly onto these dimensions, providing a robust and theoretically grounded representation of the macro-level urban form linked to navigational choices.

The variable "POI" representing the number of POIs within each grid cell, quantifies their concentration, such as shops, restaurants, schools, and hospitals within a given area. POI count serves as a proxy for functional diversity and accessibility, indicating the richness of

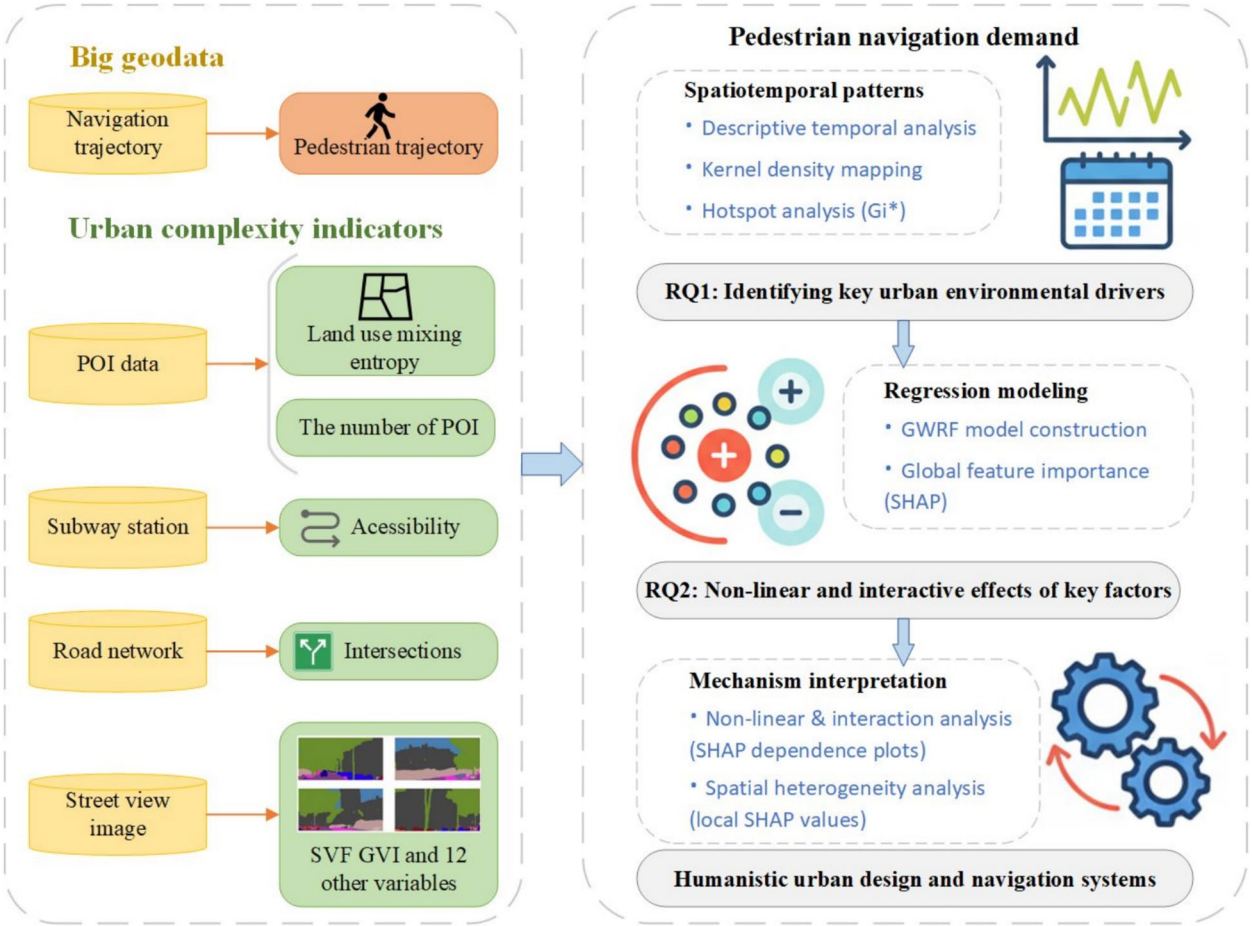


Fig. 4 The analysis framework

Table 2 Macro-scale variables

Macro-scale variable	Meaning
POI	Concentration of urban functions and amenities
Intersection	Complexity of the street network
Land use mixing entropy	Functional diversity of land use
Accessibility	Public transit service level considering supply and demand

amenities available to and convenience for residents. Higher POI counts generally suggest a more multifaceted locale with enhanced resident convenience.

The variable "Intersection" measures the number of road intersections present within each grid cell. Intersections are crucial nodes within urban transportation networks, and their counts reflect the intricacy and connectivity of the road network within a specific zone. A higher intersection count typically implies a denser road network, facilitating greater traffic flow (Xiao, D. et al., 2024). Intersection count plays a pivotal role in transportation planning, road design, and traffic flow analysis.

Land use mixing entropy, based on Shannon entropy, measures the spatial diversity of land uses within a TAZ by evaluating the distribution of POI categories. Higher entropy indicates greater functional diversity, with varied activities like shopping and leisure accessible within short distances. Lower entropy suggests uniform land use (e.g., residential or industrial), often requiring motorized transport and limiting pedestrian mobility. Despite their categorization, traditional land use datasets are coarse and lack

fine-scale functional detail. To overcome this, we aggregate POI types within each TAZ, supplemented by AOI data when needed, to calculate entropy values for each grid cell. These values reflect the urban functional layout (Fig. 5).

To mitigate potential enclosure effects within grid cells, a search radius (r = 100 m) is implemented. POIs falling within this radius, extending beyond the cell boundaries, are also incorporated into the entropy calculation for each grid cell *i*. The entropy value H_i for each grid cell *i* is then calculated as:

$$H_i = -\sum_{k=1}^{n}(p_{k,i} * lnp_{k,i}) \tag{1}$$

$$p_{k,i} = \frac{N_{k,i}}{N_{total,i}} \tag{2}$$

Here, n represents the total number of POI categories and $p_{k,i}$ denotes the proportion of category k POIs within grid cell i. $N_{k,i}$ signifies the count of category k POIs. $N_{total,i}$ represents the total number of POIs within the grid cell's buffer zone.

The final macro-scale variable is accessibility, represented by subway transit to characterize the critical interaction between public transport and pedestrian behavior. This score was calculated for each TAZ using a two-step floating catchment area method with a Gaussian distance decay function. The method determines a final accessibility score A_i by first calculating a supply-to-demand ratio for each station, following which these weighted ratios are summed for each demand origin point:

$$A_i = \Sigma_{j\epsilon\{i\ in\ service\ area\}} G(d_{ij}, d_0) R_j \tag{3}$$

Here, $G(d_{kj}, d_0)$ represents the Gaussian distance decay function and R_j reflects supply-side accessibility. The service capacity S_j that constitutes R_j was computed via Eq. 5 (Shi & Yan, 2023).

Complementing the macro-level analysis, micro-scale variables were chosen to objectively capture the physical and perceptual quality of the immediate street environment, which significantly associated with pedestrian comfort, sense of safety, and route choice behavior. This approach aligns with the visual dimension of urban complexity, emphasizing the role of the immediate scene associated with spatial perception. Metrics such as the Green View Index (GVI) and Sky View Factor (SVF) are commonly used (Zhang, A. et al., 2022). Their single-point formulas are as follows:

$$GVI_k = \frac{1}{4}\sum_{i=1}^{4}\left(\frac{A_{vegetation,i}}{A_{total,i}}\right) \times 100\% \tag{4}$$

$$SVF_k = \frac{1}{4}\sum_{i=1}^{4}\left(\frac{A_{sky,i}}{A_{total,i}}\right) \times 100\% \tag{5}$$

Here, k is the sample point index, i is the direction index, and A represents the pixel count of the suffix category within each image.

This study employs a more comprehensive suite of features for a holistic analysis. DeepLab V3 + was used to perform a pixel-wise classification of 19 distinct urban elements. From this initial set, variables with consistently low proportions across the study area and those deemed unrelated to pedestrian wayfinding were excluded, resulting in the final 14 features used for analysis. These final variables are similar to GVI and SVF. They represent the

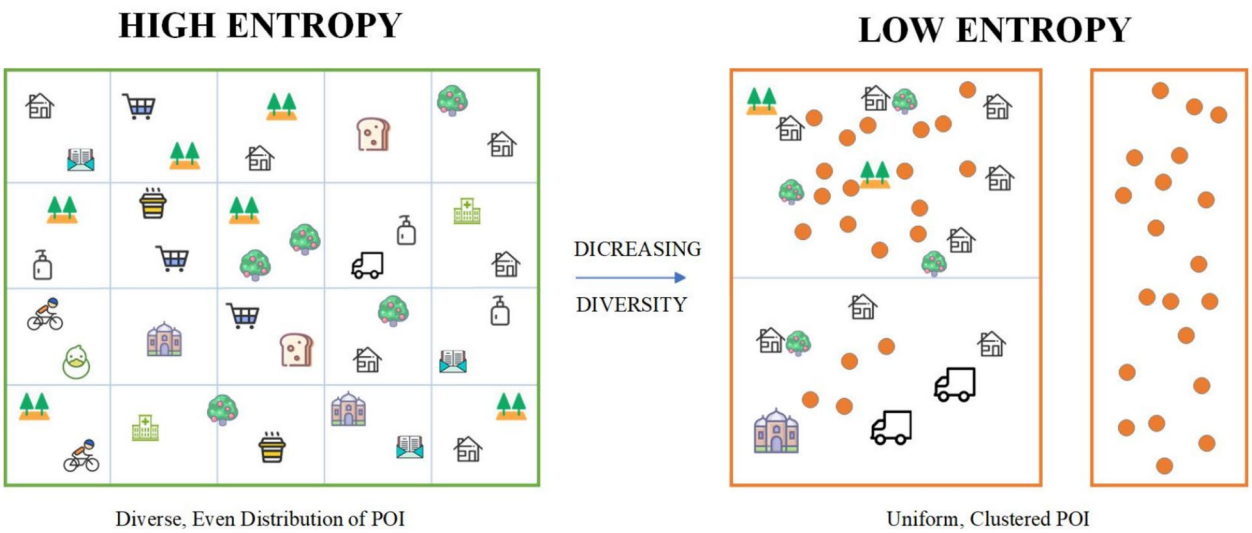


Fig. 5 Sketch map of land use factors calculation

pixel proportion of each key element, averaged across panoramic viewpoints and aggregated to the TAZ level to provide a stable, representative measure of the local visual environment. The 14 visual variables are listed in Table 3.

3.2 Analytical models

To investigate the complex and spatially varying relationships between the multi-scale urban environment and pedestrian navigation demand, this study implements a sophisticated analytical approach—the Geographically Weighted Random Forest (GWRF) model—as the core predictive tool, employing the SHapley Additive exPlanations (SHAP) method for in-depth model interpretation.

To systematically investigate the complex, and potentially non-linear and spatially varying relationships between the multi-scale urban environment and pedestrian navigation demand, this study employs a progressive modeling framework. It begins with a traditional Ordinary Least Squares (OLS) regression, serving as a global, linear baseline. It then addresses the potential for spatial heterogeneity by including a Geographically Weighted Regression (GWR) model, allowing relationships to vary locally. Next, to capture non-linear effects and high-order interactions, a global Random Forest (RF) model is incorporated into the comparison. Finally, a GWRF model is utilized to simultaneously model both non-linearity and spatial heterogeneity. This multi-model approach enables a systematic comparison of performance, thereby diagnosing the nature of the underlying relationships (i.e., linear vs. non-linear, stationary vs. non-stationary). The model that demonstrates superior explanatory power is then selected for the final, in-depth interpretation. The model comparison is detailed in Section 4.2.

A critical limitation of a standard RF model, however, is its global nature. It operates under the assumption that learned relationships are spatially stationary. In an urban context, this implies that the influence of an environmental feature—such as the number of POIs or the degree of sky visibility—is assumed to be identical across the entire study area. This assumption is often untenable in geographical analysis, as it fails to account for the spatial heterogeneity that is fundamental to understanding urban phenomena. GWRF overcomes this limitation by integrating the local regression principles of GWR, allowing the model to reveal how the influence of environmental features varies across different parts of a city, which is essential for answering our second research question.

Furthermore, we prioritized GWRF over emerging spatial deep learning alternatives, such as Spatial Graph Neural Networks (SGNNs) or Convolutional Neural Networks (CNNs), to ensure trustworthy urban analytics. Unlike deep learning models that often rely on opaque latent embeddings which are difficult to translate into policy, GWRF offers a 'structurally interpretable' framework (Fayaz & Galasso, 2024). By explicitly constructing local sub-models for each spatial unit, it preserves the observable spatial heterogeneity of urban mechanisms.

Although GWRF is a powerful predictive model, its inherent complexity still presents challenges for quantifying the specific contributions of each variable directly. To overcome this, we employ the game theory-based SHAP approach for model interpretation. It provides a unified framework to explain the output of any machine learning model. First, by calculating the average marginal contribution of each feature across all predictions, the SHAP framework quantifies the global importance of every environmental variable. This step identifies the dominant factors correlated with navigation demand and directly addresses RQ1. Second, through SHAP dependence and summary plots, we can visualize and comprehend the complex, non-linear relationships and interaction effects between variables, providing a nuanced understanding of the mechanisms that shape pedestrian navigation and contributing to the core of RQ2.The precise formula for the SHAP calculation is:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \tag{6}$$

Here, F represents the complete set of features, S denotes a subset of features excluding feature i, and $f(S)$ signifies the model's prediction value using feature subset S.

Table 3 Microscopic independent variable from SVI

SVI variable	Meaning
Person	Percentage of pedestrians in the field of view
Building	Percentage of buildings
Bicycle	Percentage of bicycles
Traffic light	Percentage of traffic lights
Sky	Sky View Factor
Traffic sign	Percentage of traffic signals
Sidewalk	Percentage of sidewalks
Pole group	Percentage of pole sets
Vegetation	Green View Index
Wall	Percentage of walls
Fence	Percentage of fences
Road	Percentage of roads
License plate	Percentage of license plates
Terrain	Percentage of terrain

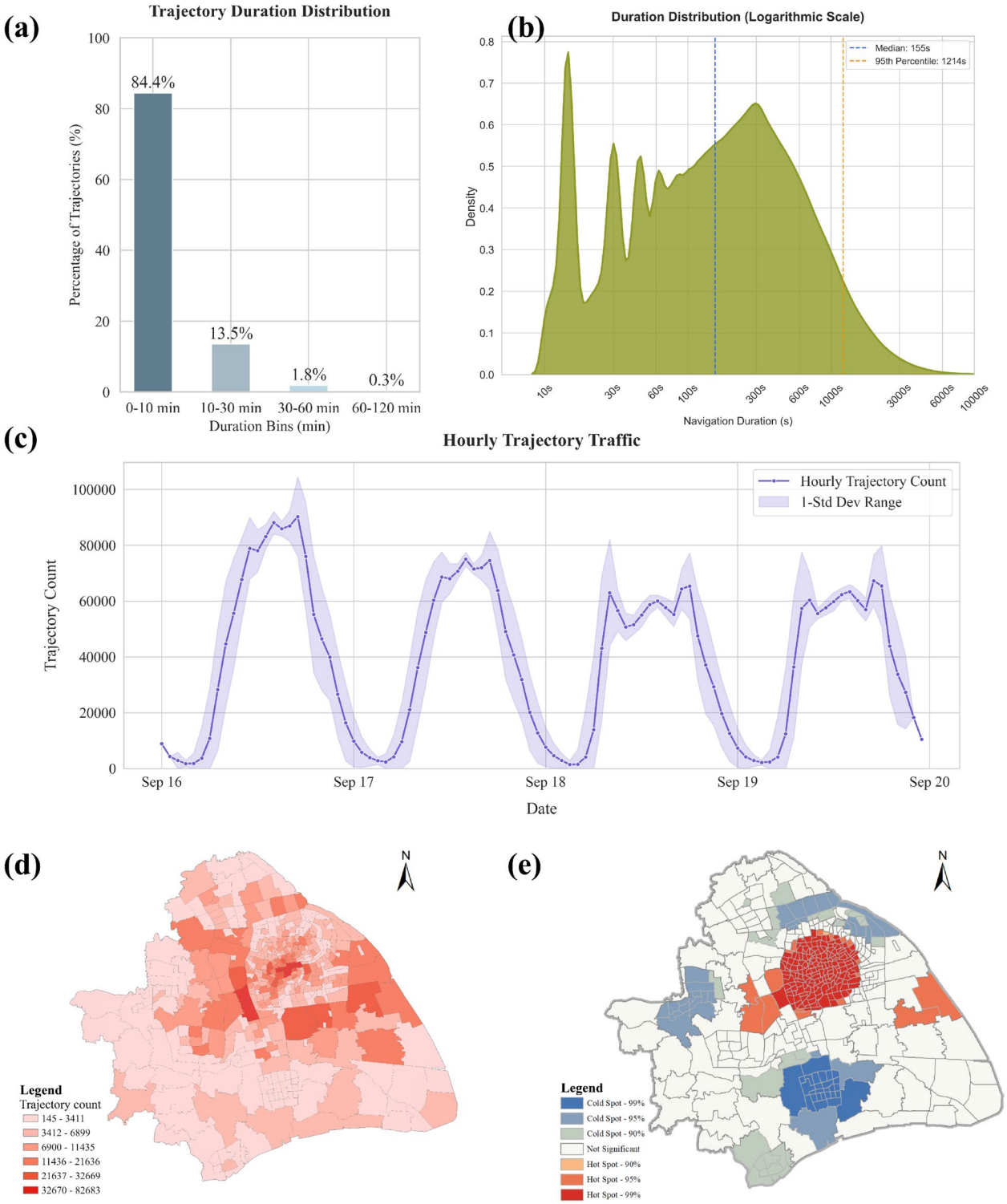


Fig. 6 Basic spatiotemporal analysis results. (**a**) Trajectory duration distribution (**b**) Trajectory duration quantile plots (**c**) Hourly traffic statistics (**d**) Trajectory point density plots (**e**) Gi* statistic for trajectories

4 Results

4.1 Spatiotemporal differentiation of trajectory

The majority of pedestrian navigation trajectories lasted between 0 and −10 min (Fig. 6a).50% of the trajectories lasted no longer than 155 s, with 95% ending within 1,214 s (Fig. 6b). This concentration of short-duration navigation events indicates that wayfinding assistance is most often sought for immediate, micro-scale navigational tasks, such as locating a nearby destination or traversing a complex local environment.

Weekend (i.e., September 16 to 17) activity was consistently higher, peaking in the morning and gradually declining thereafter (Fig. 6c). In contrast, weekday patterns exhibited moderate peaks around 08:00 and 18:00, consistent with commuting behavior, but lacked the sharp rush-hour spikes typically associated with vehicular traffic (Fig. 6c). Spatially, the highest concentrations of pedestrian activity were observed in the city center, with a gradual decline toward the periphery (Fig. 6d). Based on Gi* statistics, significant hotspots (in red) and cold spots (in blue) were identified with confidence levels ranging from 90 to 99%. Hotspots were found to be predominantly clustered in central urban areas, while cold spots were concentrated in the southern part of the city, highlighting spatial disparities in pedestrian mobility patterns (Fig. 6e).

Table 4 Performance of the different regression models

Regression model	R^2	AICc	RMSE
OLS	0.47	840.109	5302.30
GWR	0.65	723.293	3602.78
RF	0.69	2130.81	3860.08
GWRF	**0.71**	**1446.82**	**3307.92**

Note: Bold values indicate the best model performance

4.2 Identifying key urban environmental factors

To accurately capture the complex relationship between urban complexity and pedestrian navigation demand, this study compared four models (Table 4): OLS, GWR, RF, and GWRF. The results reveal that the GWRF model achieved the best performance (R2 = 0.71, RMSE = 3307.92), demonstrating superior ability in handling non-linear effects and spatial heterogeneity. It was, thus, selected as the final analytical model.

GWRF was interpreted using SHAP to identify the key environmental correlates. POI, accessibility, and person emerge as the most influential predictors, followed by intersection, land-use entropy, bicycle infrastructure, traffic sign, building, traffic light, and sky visibility (Fig. 7a). Macro-scale infrastructure variables—such as accessibility and intersection—show stronger associations than micro-scale features (Fig. 7a). Among the top 10 features, POI count exhibits the widest SHAP value

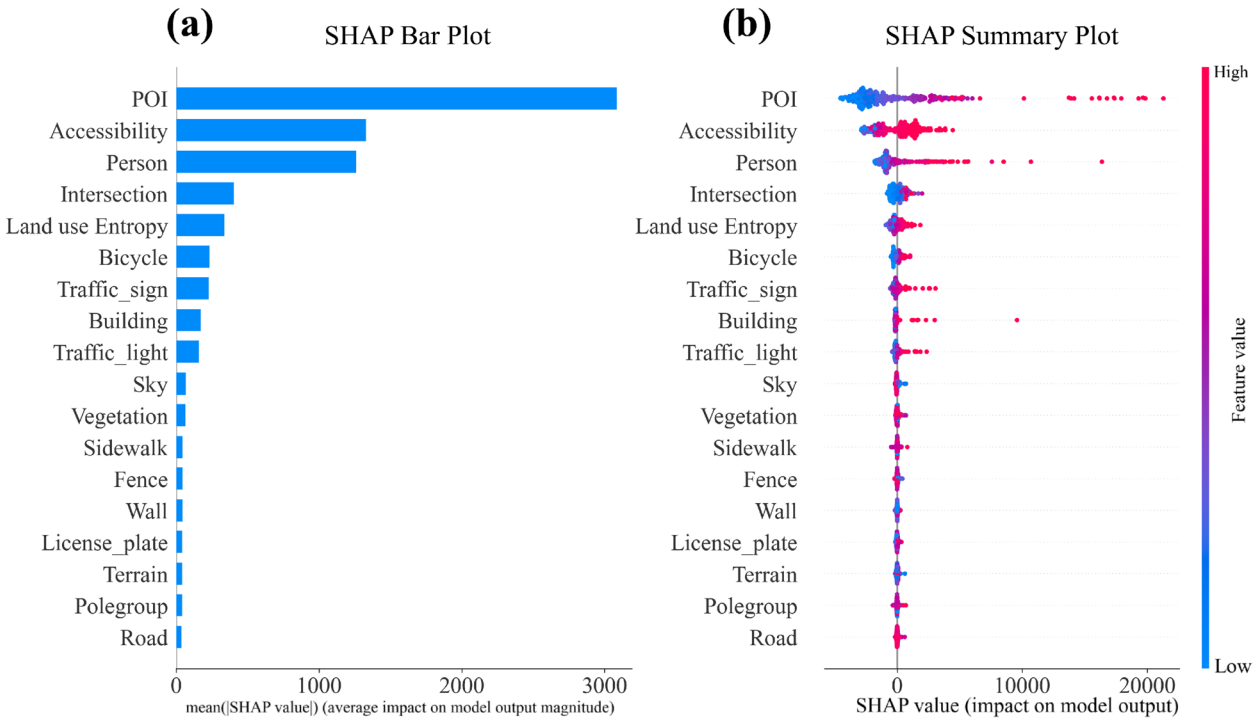


Fig. 7 Plot of overall nonlinear relationship results (**a**) SHAP global analysis; (**b**) SHAP local interpretation

distribution, suggesting both positive and negative contributions to the predicted number of navigation trajectories, although predominantly exerting a positive effect (Fig. 7b). Accessibility and person also display high variability and strong positive impacts on model predictions (Fig. 7b). In contrast, other variables demonstrate more concentrated SHAP distributions, indicating more consistent and narrowly distributed effects (Fig. 7b).

To examine the specific role of population scale, we conducted a population-normalization test (Supplementary Material S2, Fig. S1). The results show that while the coefficient of POI counts decreased significantly, structural variables (Accessibility, Intersection) and visual features (Traffic signs) maintained their statistical significance and high relative importance. Additionally, sensitivity analyses excluding POI and Person variables (Supplementary Material S4) yielded a model with an R2 of 0.56, where environmental physical features remained significant predictors. Finally, the temporal comparison between weekday and weekend models (Supplementary Material S3) revealed consistent coefficient patterns across core environmental features, with no significant deviation in variable rankings observed between the two periods.

4.3 Non-linear and interactive effects of key factors

The SHAP dependence plots in Fig. 8 reveal that environmental features relate to navigation demand through several distinct and complex non-linear patterns. A primary pattern is the saturation effect. For instance, the positive association of POI (Fig. 8a) and intersection (Fig. 8d) counts on navigation demand increases sharply as their values rise from zero. However, as these values reach higher ranges, the rate of increase gradually flattens, suggesting a potential plateau effect. In sharp contrast, traffic sign (Fig. 8e), traffic light (Fig. 8i), and t person (Fig. 8c) exhibit an accelerating positive effect. At lower levels of these features, their impact is weak; but, beyond this point, the growth rate of their positive association continuously increases, resembling an exponential-like trend. Furthermore, accessibility (Fig. 8b) displays a clear U-shaped relationship: within its lower-to-moderate range, it is negatively associated with navigation demand (indicated by a negative slope); after this point, its effect rapidly reverses into a strong positive correlate (indicated by a positive slope).

In contrast to the predominantly positive associations exhibited by most of the aforementioned variables, only sky visibility (Fig. 8j) reveals a clear and consistent negative linear relationship. Other variables follow more stable trends: for example, building (Fig. 8h) and bicycle (Fig. 8g) are associated with stable positive linear trends. Finally, a subset of features, including vegetation (Fig. 8p) and road (Fig. 8n), reveal weak or ambiguous relationships, with their SHAP values clustering around zero, indicating a negligible impact.

Several of the most significant relationships are highlighted in the analysis of interaction effects (Fig. 9). These interactions predominantly revolve around the POI count and person, identifying them as key factors that modulate the influence of other environmental features.

The number of POIs emerges as a primary modulating factor for a wide range of environmental features (Fig. 9a-f). For variables with a generally positive impact—such as traffic light, person, accessibility, land use entropy, and bicycle—a high POI count, indicated by red points on the color scale, consistently amplifies their positive SHAP values, demonstrating a stronger positive association with navigation demand when these features co-occur in POI-rich areas. Conversely, POI count mitigates the negative impact of sky visibility (Fig. 9e). In areas with high POI counts, the negative SHAP values associated with sky are smaller (closer to zero), whereas in areas with low POI counts (blue points), the negative SHAP values are larger.

A similar synergistic effect is observed with the presence of other pedestrians (person) as a modulating variable (Fig. 9g-i). High person values enhance the positive SHAP values of traffic sign, intersection, and POI, indicating that the positive relationship between these features and navigation demand is more pronounced in crowded settings.

4.4 Spatial heterogeneity of key factor impacts

The spatial distribution maps of SHAP values from the GWRF model reveal that the associations of key environmental factors with navigation demand are not uniform; instead, they exhibit significant spatial heterogeneity with distinct patterns for different variables (Fig. 10). In all maps, the light-yellow areas represent a neutral zone where SHAP values are close to zero, serving as a watershed between the positive, positive associations (warm tones) and the negative, negative associations (cool tones). The impact of POI (Fig. 10a) follows a centrally enhanced pattern with localized inversions. Its positive driving effect is dominant across the study area, peaking in the core urban districts. However, this positive effect is not absolute, as distinct zones of negative impact exist in some central and southern locations. In contrast, the impact of traffic sign (Fig. 10b) displays a highly heterogeneous and scattered pattern. Unlike the clear core-periphery structure of POI, strong positive and negative impact zones coexist as patches distributed throughout the city without a regular spatial concentration. Finally, the impact of sky visibility (Fig. 10c) exhibits a pattern of central suppression with peripheral inversion. Its

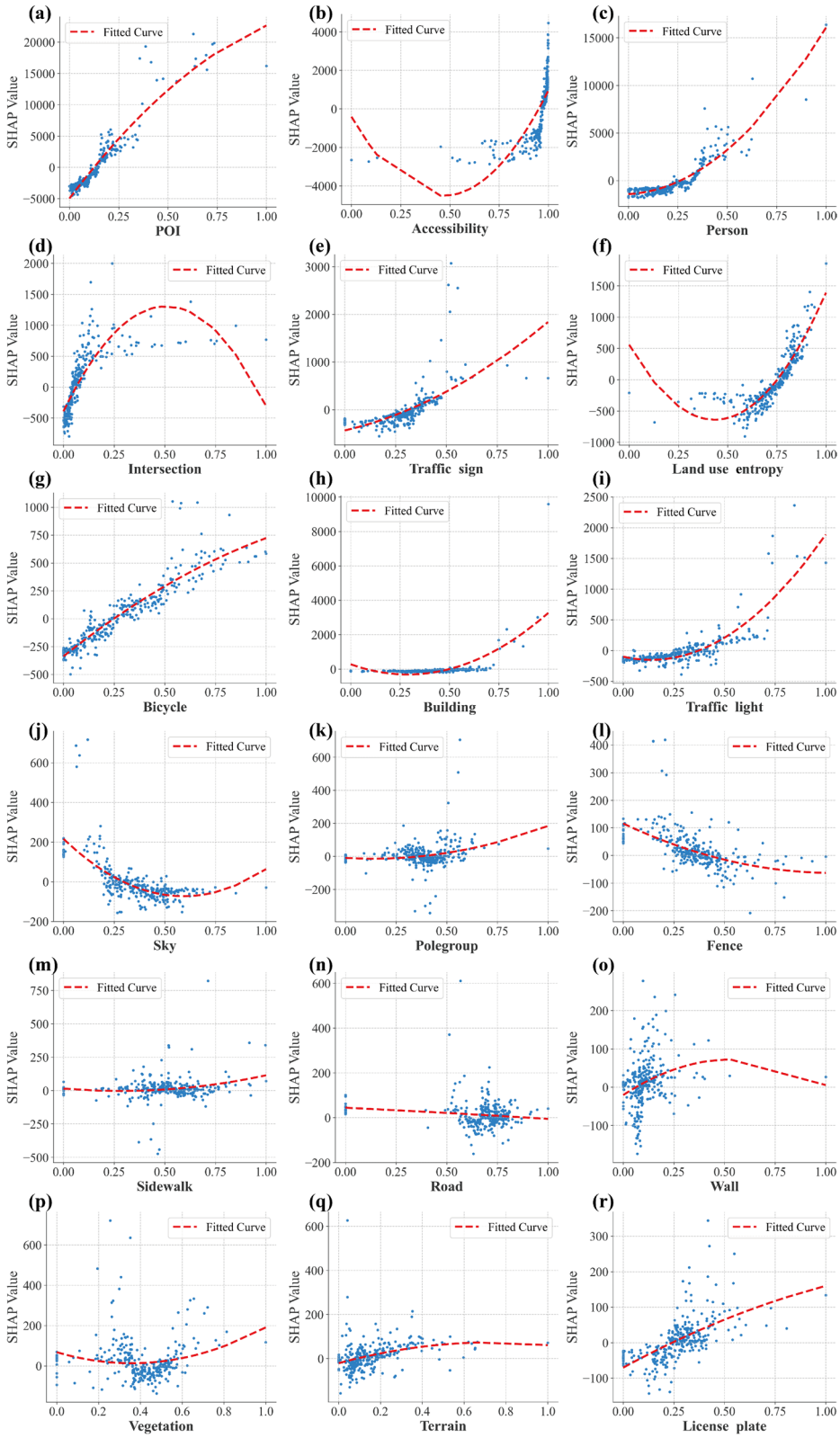


Fig. 8 Non-linear effects of individual urban features (**a**) POI; (**b**) Accessibility; (**c**) Person; (**d**) Intersection; (**e**) Traffic sign; (**f**) Land use entropy; (**g**) Bicycle; (**h**) Building; (**i**) Traffic light; (**j**) Sky; (**k**) Polegroup; (**l**) Fence; (**m**) Sidewalk; (**n**) Road; (**o**) Wall; (**p**) Vegetation; (**q**) Terrain; (**r**) License plate

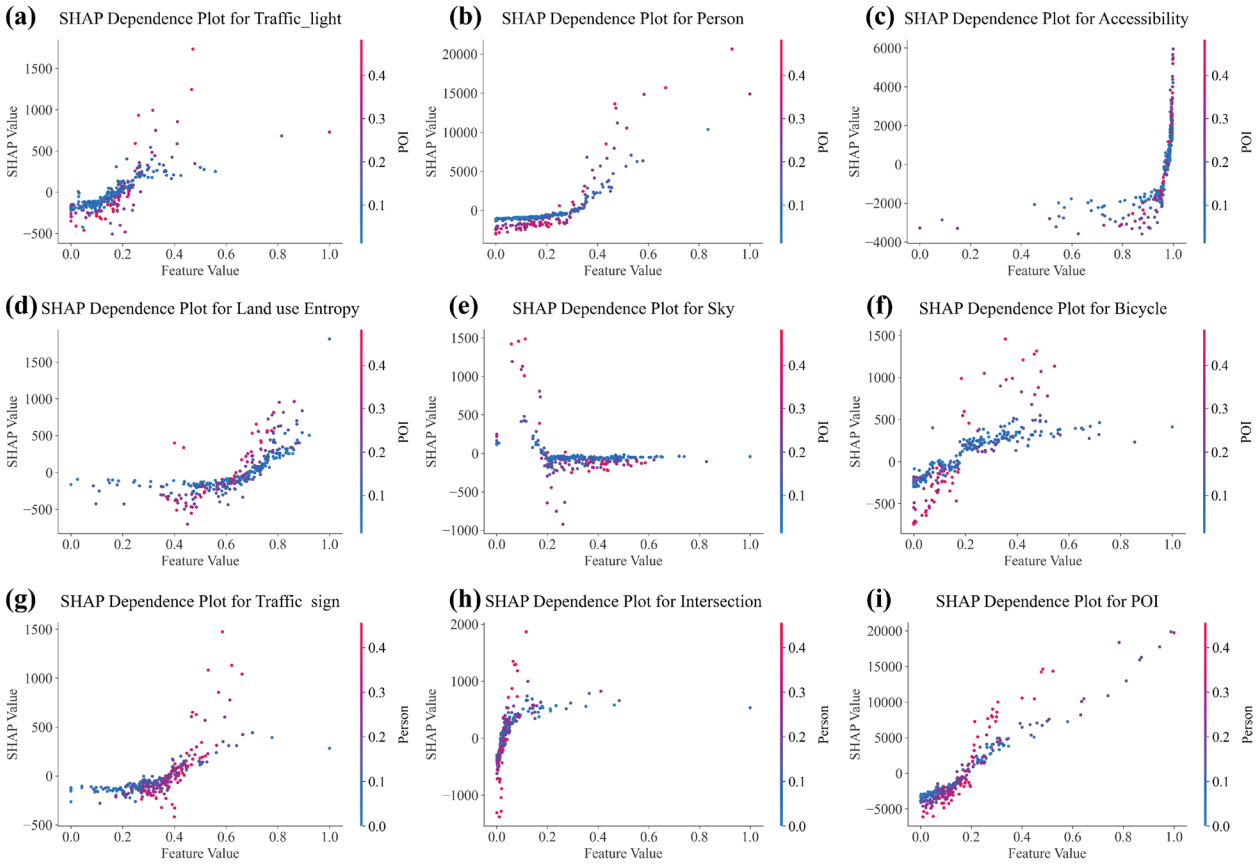


Fig. 9 Interaction effects between urban features (**a**) Traffic light; (**b**) Person; (**c**) Accessibility; (**d**) Land use Entropy; (**e**) Sky; (**f**) Bicycle; (**g**) Traffic sign; (**h**) Intersection; (**i**) POI

predominantly negative effect on navigation demand is most pronounced in the dense city center. This relationship, however, inverts in specific eastern and peripheral zones, where sky visibility has a positive driving effect.

5 Discussion

5.1 Synergy of macro-structure and micro-perception

This study sought to identify the primary environmental features related to pedestrian navigation demand (RQ1). The findings reveal that a combination of macro-structural and micro-perceptual factors are synergistically associated with the perceived complexity of the urban environment. The analysis identified the number of POIs, accessibility, presence of other people, and intersection count as the most dominant correlates of navigation behavior. This empirically supports our core hypothesis that the intensity of navigation trajectories serves as a quantifiable proxy for "perceived navigational complexity"—the cognitive load required for wayfinding in an environment.

The prominence of macro-scale variables aligns with foundational urban theories. High densities of POIs and intersections—key components of the "5Ds" framework—relate to a higher number of decision points and potential destinations a pedestrian must process. This informational overload can disrupt the formation of clear "cognitive maps" and reduce a city's "legibility" (Ibrahim et al., 2024). When the environment's structure is too intricate, individuals are more likely to experience disorientation and spatial anxiety, leading to greater reliance on navigational aids (Nori et al., 2023). Similarly, high accessibility, despite being beneficial for transit, can create complex multimodal hubs where navigating the last mile on foot becomes a significant cognitive challenge, as reflected in its strong positive association with navigation demand (Arnold et al., 2023).

Although macro-structures form the backbone of urban complexity, micro-scale perceptual features add a critical contextual layer that traditional models often miss. Our results reveal that eye-level elements such as the proportion of pedestrians, traffic signs, buildings, and sky visibility are also significant predictors. For instance, a high percentage of buildings and low sky visibility create a sense of enclosure, obstructing long-distance

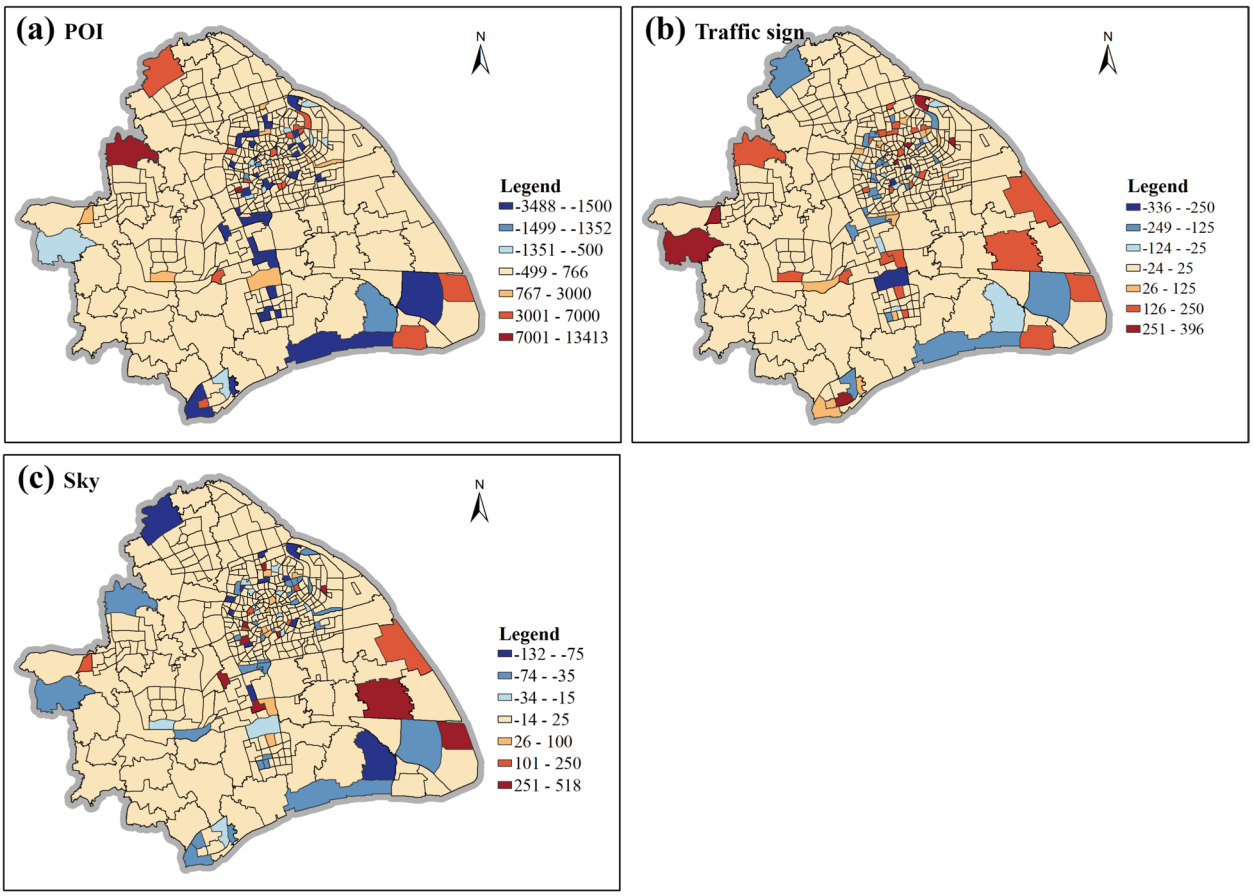


Fig. 10 **(a)** SHAP value spatial distribution (POI); (**b**) SHAP value spatial distribution (Traffic sign); (**c**) SHAP value spatial distribution (Sky)

sightlines and natural orientation cues, thereby increasing spatial disorientation. Beyond simply integrating variables, this study contributes to the refinement of such frameworks. By ranking the importance of variables, our results demonstrate that although some micro-scale features are critical, others—including fence, pole group, and terrain—have minimal predictive power. This suggests that future models could be streamlined by excluding these low-impact variables, creating a more efficient and targeted framework for assessing urban complexity.

Finally, our robustness checks (Supplementary Material S2) clarify the distinct nature of these associations by bridging the macro and micro dimensions. While the baseline model confirms that POI counts are the primary correlate of total navigation volume, the population-normalized model reveals a critical shift in dominance: when the scale effect of crowdedness is removed, the relative importance of POI diminishes, whereas both structural features (e.g., Accessibility) and visual features (e.g., Traffic signs) exhibit sustained or increased significance. This distinction empirically suggests that navigation demand is not merely a byproduct of commercial concentration or crowd density. Instead, it is characterized by a dual mechanism: where high POI counts are associated with the potential movement flow, while the intrinsic environmental complexity—encompassing both macro-spatial structure and micro-visual legibility—reflects the actual cognitive cost of wayfinding.

5.2 Implications for human-centric urban design

Beyond identifying the factors that matter, this research delves into how and where these features are associated with navigation demand, uncovering the complex mechanisms essential to formulate human-centric urban policy (RQ2). The findings demonstrate that the relationships between environmental features and navigation behavior are profoundly non-linear, interactive, and spatially heterogeneous.

The non-linear effects reveal that simple and linear assumptions are inadequate for urban planning (Yang, L. et al., 2024a). For instance, the influence of POI count exhibits a diminishing return, suggesting a phase of cognitive saturation, where additional stimuli correspond to a diminishing marginal change in navigational demand.

Conversely, accessibility displays a U-shaped relationship, with its impact being suppressive at the lowest levels before rising sharply to become a powerful positive correlate at high levels. This indicates that low-accessibility areas, because of their structural simplicity, require less navigation, whereas high-accessibility areas—such as dense multimodal hubs—are inherently complex and significantly are associated with a higher need for wayfinding assistance. These nuanced patterns provide critical evidence to understand the non-linear dynamics of urban configuration and connectivity in urban design.

Crucially, the analysis of interaction effects provides the strongest evidence for a context-aware, human-centric approach, revealing that the impact of any single environmental feature is not static but conditioned by its surrounding context. High POI count does not merely add an independent layer of complexity; it is associated with greater navigational challenges when encountering intersections. This finding implies that planning dense commercial districts necessitates special attention to the legibility of internal nodes. Furthermore, crowded environments significantly increase the navigation demand associated with intersections and traffic signs. Thus, urban spaces must be evaluated for their functional performance under the dynamic pressure of human flows, suggesting that high-traffic pedestrian zones may require clearer and more redundant wayfinding systems.

Sensitivity testing (Supplementary Material S4) further supports the robust association between urban form and navigation. Even without the potential amplifying influence of high POI counts or crowded environments, street network structures and visual markers remain significantly linked to pedestrian cognitive load. This reinforces the finding that while a high concentration of POIs appears to exacerbate wayfinding challenges, the intrinsic complexity of the environment itself remains a primary associate of navigation demand. Finally, these interaction effects offer a blueprint to develop more sophisticated, context-aware navigation tools. A compelling example is the mitigating effect of POI density on the negative impact of low sky visibility, which suggests the potential for a navigation system to detect when a user enters a “street canyon” and proactively provide richer data on surrounding landmarks as alternative orientation cues.

Furthermore, the pronounced spatial heterogeneity of these impacts invalidates a "one-size-fits-all" approach to urban design, reflecting the city’s diverse urban fabric. Although the influence of POI count is strongest in the urban core, this effect may be reversed in certain hyper-commercialized nodes, likely because these areas contain famous landmarks that visitors already know how to find, thus reducing their reliance on map applications. The impact of traffic sign visibility is highly scattered, but the model effectively identifies "hotspots" where navigation demand is unusually high. Such locations could signal to city planners that existing signage is outdated, confusing, or insufficient, marking them as priorities for on-site inspection.

The effect of sky visibility reveals a pattern of central suppression and peripheral inversion; its suppressive impact is most pronounced in the dense city center because of the "street canyon" effect, although the relationship can reverse in suburban areas, perhaps because the open environment—combined with natural and recreational attractions—encourages people to explore more freely rather than relying on navigation. These distinct spatial patterns are rooted in the city’s evolving urban morphology and functional specialization (Durst et al., 2024). The urban core, characterized by dense historic street networks and mixed-use functionality, presents varying cognitive challenges compared to the sprawling, function-segregated suburban developments.

Finally, the robustness of these findings is further corroborated by the temporal consistency observed in our comparative models (Fig. S2). Despite the distinct behavioral motivations typically associated with weekdays (commuting, high familiarity) versus weekends (leisure, exploration), the association between navigation demand and core environmental features remains remarkably stable. This temporal consistency suggests that the cognitive challenges imposed by urban complexity are not limited to specific travel purposes. Instead, they reflect a robust spatial pattern, indicating that design interventions targeting these environmental elements could offer generalized benefits for diverse pedestrian groups.

5.3 Limitations and future prospects

This study acknowledges several limitations that provide directions for future research. Primarily, as the analysis relies on a cross-sectional dataset from a specific period in 2017, the results may not capture long-term behavioral dynamics. Although the selected dates had stable weather conditions to minimize confounding effects, future longitudinal studies should explicitly control for temporal and meteorological variations (Dong et al., 2022). while Shanghai serves as a representative case, the generalizability of these findings also warrants validation through comparative studies across cities with diverse urban forms. Methodologically, the use of TAZs as the spatial unit introduces potential Modifiable Areal Unit Problems (MAUP) (Ye & Rogerson, 2021), as macro-scale aggregation may obscure fine-grained variations in pedestrian perception. To mitigate this, we implemented high-density SVI sampling (averaging 452 points per TAZ) to ensure that the aggregated metrics remain representative of local streetscape conditions. Building on

this, future research could adopt a multi-scale strategy by applying finer spatial units—such as 100 m × 100 m grid cells—specifically in targeted regions characterized by high data density or critical spatial significance.

Furthermore, the omission of individual demographic information limits our ability to assess differential impacts across population subgroups. High environmental complexity may impose disproportionately heavier cognitive burdens on vulnerable populations: older adults experience age-related declines in spatial processing speed and working memory (Xu et al., 2024), while visually impaired pedestrians rely on non-visual sensory cues that are not captured by vision-based SVI metrics. Similarly, tourists and newcomers likely exhibit heightened navigation dependence compared to long-term residents familiar with local spatial layouts. These differential vulnerabilities have critical implications for inclusive urban design policy. Future research should stratify analyses by age, disability status, and residency duration to inform targeted interventions—such as enhanced wayfinding signage in high-complexity areas, tactile navigation aids for visually impaired users, and multilingual information systems for international visitors (Hegarty et al., 2023). Finally, navigation demand serves as a proxy for behavioral reliance on wayfinding tools rather than a direct measure of subjective disorientation. High navigation usage typically reflects habitual app dependence or destination confirmation, rather than genuine wayfinding difficulty (Lohmeier et al., 2025). To disentangle these motivations, future research integrating multi-source data—combining large-scale trajectories with targeted surveys, eye-tracking studies, or experimental wayfinding tasks—could provide richer insights into the subjective experience of urban navigation and the cognitive mechanisms underlying tool dependence.

Despite these limitations, this research provides a robust framework and actionable insights for human-centered urban planning, laying the groundwork for future investigations into the intricate relationship between urban complexity and human mobility. This study posits that while POI counts primarily account for the total volume of navigation demand, structural and visual features—specifically intersections, accessibility, and signage—exhibit a persistent association with wayfinding reliance that remains significant even after controlling for population scale. This distinction suggests that clearer signage and more accessible intersections may reduce navigation dependence. Furthermore, creating green spaces within densely populated areas can mitigate spatial disorientation (Yang, W. et al., 2024b). The observed interactions between POIs and sky visibility also offer opportunities for developing context-aware navigation tools that proactively assist users in complex "street canyons". By addressing these multi-dimensional factors, planners can create more legible, adaptive, and pedestrian-friendly urban environments.

6 Conclusion

This study advances the understanding of urban complexity and pedestrian navigation by establishing a rigorous, data-driven framework that integrates large-scale behavioral trajectories with multi-scale environmental metrics. We demonstrate that navigation demand—further corroborated by population normalization, sensitivity tests, and temporal comparisons—serves as a robust proxy for perceived environmental complexity. Our analysis reveals that macro-structural features (the POI count, accessibility, intersection count) and micro-perceptual elements (proportion of pedestrians, traffic signs) are synergistically associated with navigation behavior through profoundly non-linear and spatially heterogeneous patterns. These findings challenge 'one-size-fits-all' planning approaches and provide empirical insights for creating more legible, context-aware, and human-centered urban environments. Future research should extend this framework through longitudinal analyses, cross-city comparisons, and demographic stratification to inform inclusive urban design policies.

Supplementary Information

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Supplementary Material 1.

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Authors' contributions
Haoran Fang: Methodology, Formal analysis, and Writing Original Draft. Wei Huang: Conceptualization, Supervision, Funding acquisition, Writing Original Draft, and Review & Editing; Hangbin Wu: Supervision and Review & Editing.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Competing interests
We have no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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